



Lavanya R 250701385-2025-CSE ▾

L2**Started on** Friday, 7 August 2026, 8:46 AM**State** Finished**Completed on** Friday, 7 August 2026, 9:10 AM**Time taken** 23 mins 29 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main(){
3      int n;
4      scanf("%d",&n);
5      int count=0;
6      int i= 1;
7      count++;
8
9      int s =1;
10     count++;
11     while(s <= n)
12     {
13         count++;
14         i++;
15         count++;
16         s += i;
17         count++;
18     }
19     count++;
20     printf("%d",count);
    }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Lavanya R 250701385-2025-CSE ▾

L2

Started on	Friday, 7 August 2026, 9:11 AM
State	Finished
Completed on	Monday, 10 August 2026, 10:24 PM
Time taken	3 days 13 hours
Marks	1.00/1.00

Grade 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2  int main(){
3      int n;
4      scanf("%d",&n);
5      int count=0;
6      if(n==1)
7      {
8          count++;
9          printf("*");
10     }
11     else
12     {
13         count++;
14         for(int i=1; i<=n; i++){
15             count++;
16             for(int j=1; j<=n; j++){
17                 count++;
18                 count++;
19                 count++;
20                 break;
21             }
22             count++;
23         }
24         count++;
25     }
26     printf("%d",count);
27 }
28
29 }
```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Lavanya R 250701385-2025-CSE ▾

L2**Started on** Monday, 10 August 2026, 10:24 PM**State** Finished**Completed on** Monday, 10 August 2026, 10:29 PM**Time taken** 4 mins 33 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int main(){
3      int num,i,count=0;
4      scanf("%d",&num);
5      count++;
6      for (i = 1; i <= num; ++i)
7      {
8          count++;
9          if (num % i == 0)
10         {
11             count++;
12         }
13         count++;
14     }
15     printf("%d",count);
16 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Lavanya R 250701385-2025-CSE ▾

L2**Started on** Monday, 10 August 2026, 10:29 PM**State** Finished**Completed on** Monday, 10 August 2026, 10:40 PM**Time taken** 11 mins 3 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int main()
3  {
4      int n,c= 0;
5      scanf("%d",&n);
6      c++;
7      for(int i=n/2; i<n; i++){
8          c++;
9          for(int j=1; j<n; j = 2 * j){
10             c++;
11             for(int k=1; k<n; k = k * 2){
12                 c++;
13                 c++;
14             }
15             c++;
16         }
17         c++;
18     }
19     c++;
20     printf("%d",c);
21 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Lavanya R 250701385-2025-CSE ▾

L2**Started on** Monday, 10 August 2026, 10:40 PM**State** Finished**Completed on** Monday, 10 August 2026, 10:46 PM**Time taken** 5 mins 52 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  void reverse(int n)
3  {
4      int rev = 0, remainder;
5      int count=2;
6      while (n != 0)
7      {
8          count++;
9          remainder = n % 10;
10         count++;
11         rev = rev * 10 + remainder;
12         count++;
13         n/= 10;
14         count++;
15     }
16     count++;
17     printf("%d",count);
18 }
19
20 int main(){
21     int n;
22     scanf("%d",&n);
23     reverse(n);
24 }
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)